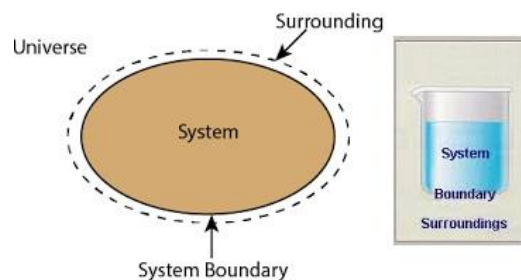


THERMODYNAMICS

Thermodynamic System

A thermodynamic system is a quantity of matter of fixed identity, around which we can draw a boundary. The boundaries may be fixed or moveable. Work or heat can be transferred across the system or boundary.

Examples: Refrigerator, air-conditioner, washing machine, heat exchange, a utensil with hot water etc.

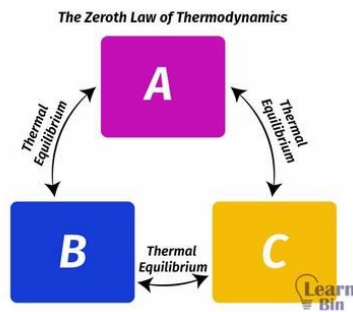


Open system	Closed system	Isolated system
The system in which the transfer of mass as well as energy can take place across its boundary is called as an open system.	The system in which the transfer of energy takes place across its boundary with the surrounding, but no transfer of mass takes place is called as closed system. The closed system is fixed mass system.	The system in which neither the transfer of mass nor that of energy takes place across its boundary with the surroundings is called as isolated system. Here there will neither transfer of mass nor that of energy.
Example: Open system is boiling water in an open vessel, where transfer of heat as well as mass in the form of steam takes place between the vessel and surrounding.	Example: The water being heated in the closed vessel, when water will get heated but its mass will remain same.	Hot water, coffee or tea kept in the thermos flask is closed system.
		

Zeroth Law of Thermodynamics

- Zeroth law of thermodynamics is one of the four laws of thermodynamics. The credit for formulating the law goes to Ralph H. Fowler. Interestingly, the zeroth law of thermodynamics was actually developed much later than the original three laws. However, there was some confusion regarding the nomenclature, whether it should be named the fourth law or some other name. The complication arose because the new law gave a much clearer definition of the temperature and basically replaced what the other three laws had to state. Fowler finally came up with the name to end this conflict.
- The zeroth law of thermodynamics frames an idea of temperature as an indicator of thermal equilibrium.
- Thermal Equilibrium:** Temperature is a property that distinguishes thermodynamics from other sciences. This property can distinguish between hot and cold. When two or more bodies at different temperatures are brought into contact then after some time they attain a common temperature and they are said to exist in thermal equilibrium. Systems are said to be in thermal equilibrium if there is no heat transfer, even if they are in a position to transfer heat, based on other factors. For example, if we put food in the refrigerator overnight then that food is in thermal equilibrium with the air of that refrigerator. Heat no longer flows from food to the air or from the air to the food, this state is known as

Zeroth Law of Thermodynamics



Statement:

“The zeroth law of thermodynamics states that if two thermodynamic systems are each in thermal equilibrium with a third, then the two systems are in thermal equilibrium with each other also”.

When a body 'A' is in thermal equilibrium with another body 'b', and also separately in thermal equilibrium with a body 'C', then body 'B' and 'C' will also be in thermal equilibrium with each other. There are also various ways to state the zeroth law of thermodynamics. However, in simple terms, it can be said, “Systems that are in thermal equilibrium exist at the same temperature”.

First Law of Thermodynamics

- Before we get into the first law of thermodynamics we need to understand the relation between heat and work and the concept of internal energy. Just like mass, energy is always conserved i.e. it can neither be created nor destroyed but it can be transformed from one form to another.
- **Internal energy** is a thermodynamic property of the system that refers to the energy associated with the molecules of the system which includes kinetic energy (due to translational, rotational and vibrational motion of molecules which only depends on T) and potential energy (due to intermolecular force) which depends on the separation between the molecules.
- Whenever a system goes through any change due to interaction of heat, work and internal energy, it is followed by numerous energy transfer and conversions. However, during these transfers, there is no net change in the total energy. So, the first law of thermodynamics affirms that **heat is a form of energy**. The first law of thermodynamics tells that the energy of the universe remains the same. Though it may be exchanged between the system and the surroundings, it can't be created or destroyed. The law basically relates to the changes in energy states due to work and heat transfer. It redefines the **conservation of energy** concept.

First Law of thermodynamics

Statement: “ If some heat is supplied to a system which can do work, then the quantity of heat absorbed by the system is equal to the sum of increase in internal energy of the system and the external work done by the system”.

Explanation: If dQ be the amount of heat absorbed by a system and dU be the change in internal energy and dW is the work done by the system due to absorption of heat, then according to the 1st law of thermodynamics,

$$dQ = dU + dW$$

Specific Heat (S): The specific heat of material is defined as the quantity of heat required to raise the temperature of unit 1 gm of that material through 1 degree Celsius.

$$\text{Specific heat, } S = \frac{Q}{m\Delta T}$$

The unit of specific heat is in C.G.S. is $\text{cal g}^{-1}\text{C}^{-1}$ and in S.I. is $\text{JK g}^{-1}\text{K}^{-1}$.

Molar Specific Heat at Constant Volume (C_v): It is defined as the amount of heat energy required to raise the temperature of one mole of a gas by 1°C at **constant volume**. It is represented by C_v and given by,

$$C_v = \left(\frac{dQ}{n dT} \right)_v$$

Molar Specific Heat at Constant Pressure (C_p): It is defined as the amount of heat energy required to raise the temperature of one mole of a gas by 1°C at **constant pressure**. It is represented by C_p and given by,

$$C_p = \left(\frac{dQ}{n dT} \right)_p$$

(iii) Ratio of Specific Heats for Different Gases (Atomicity of gases):

(a) Mono-atomic Gases:

Helium and argon are mono-atomic gases. The mono-atomic gases have only translational motion; hence they have three translational degrees of freedom. The average energy of the molecule at temperature T is given by

$$E_{Trans} = \frac{1}{2}mu^2 + \frac{1}{2}mv^2 + \frac{1}{2}mw^2$$

By the law of equipartition of energy we have, $\frac{1}{2}mu^2 = \frac{1}{2}mv^2 = \frac{1}{2}mw^2 = \frac{1}{2}kT$.

Hence, $E_{Trans} = \frac{1}{2}kT + \frac{1}{2}kT + \frac{1}{2}kT = \frac{3}{2}kT$

Thus the energy per mole of the gas is given by, $U = N \times \frac{3}{2}kT = \frac{3}{2}RT$ [$Nk = R$]

Hence, $\frac{dU}{dT} = \frac{3}{2}R = C_V$ [Since, $C_V = \frac{dU}{dT}$]

Again we know,

$$C_P - C_V = R$$

$$\text{Or, } C_P - \frac{3}{2}R = R$$

$$\therefore C_P = \frac{5}{2}R$$

So the ratio of specific heats, $\gamma = \frac{C_P}{C_V} = \frac{\frac{5}{2}R}{\frac{3}{2}R} = 1.67$

Thus the ratio of specific heat capacities of mono-atomic gas is 1.67.

(b) Diatomic Gases:

Di-nitrogen, di-oxygen, and di-hydrogen are diatomic gases. The diatomic gases have translational motion (translational degrees of freedom) as well as rotational motion (rotational degree of freedom). The average energy of the molecule at temperature T is given by

$$E_T = E_{Trans} + E_{Rot}$$

$$\text{Or, } E_T = \frac{1}{2}mu^2 + \frac{1}{2}mv^2 + \frac{1}{2}mw^2 + \frac{1}{2}I\omega_1^2 + \frac{1}{2}I\omega_2^2$$

From equipartition law of energy, $\frac{1}{2}mu^2 = \frac{1}{2}mv^2 = \frac{1}{2}mw^2 = \frac{1}{2}I\omega_1^2 = \frac{1}{2}I\omega_2^2 = \frac{1}{2}kT$.

$$\text{Hence, } E_T = \frac{1}{2}kT + \frac{1}{2}kT + \frac{1}{2}kT + \frac{1}{2}kT + \frac{1}{2}kT = \frac{5}{2}kT$$

Thus the energy per mole of the gas is given by, $U = N \times \frac{5}{2}kT = \frac{5}{2}RT$ [$Nk = R$]

$$\text{Hence, } \frac{dU}{dT} = \frac{5}{2}R = C_V \quad [\text{Since, } C_V = \frac{dU}{dT}]$$

Again we know,

$$C_P - C_V = R$$

$$\text{Or, } C_P - \frac{5}{2}R = R$$

$$\therefore C_P = \frac{7}{2}R$$

$$\text{So the ratio of specific heats, } \gamma = \frac{C_P}{C_V} = \frac{\frac{7}{2}R}{\frac{5}{2}R} = 1.40$$

Thus the ratio of specific heat capacities of diatomic gas is 1.40.

(c) Tri-atomic Gas:

Tri-oxygen (ozone) and carbon-dioxide are tri-atomic gases. The tri-atomic gases have translational motion, rotational motion as well as vibrational motion, hence has three translational degrees of freedom and two rotational degrees of freedom. For non-rigid molecules, there is an additional vibrational motion.

The average energy of the molecule at temperature T is given by

$$E_T = E_{Trans} + E_{Rot} + E_{Vib}$$

$$\text{Or, } E_T = \frac{1}{2}mu^2 + \frac{1}{2}mv^2 + \frac{1}{2}mw^2 + \frac{1}{2}I\omega_1^2 + \frac{1}{2}I\omega_2^2 + \frac{1}{2}ky^2$$

From equipartition law of energy we have,

$$\frac{1}{2}mu^2 = \frac{1}{2}mv^2 = \frac{1}{2}mw^2 = \frac{1}{2}I\omega_1^2 = \frac{1}{2}I\omega_2^2 = \frac{1}{2}ky^2 = \frac{1}{2}kT$$

$$\text{Hence, } E_T = \frac{1}{2}kT + \frac{1}{2}kT + \frac{1}{2}kT + \frac{1}{2}kT + \frac{1}{2}kT + \frac{1}{2}kT = 3kT$$

Thus the energy per mole of the gas is given by, $U = N \times 3kT = 3RT$ [$Nk = R$]

$$\text{Hence, } \frac{dU}{dT} = 3R = C_V \quad [\text{Since, } C_V = \frac{dU}{dT}]$$

Again we know,

$$C_P - C_V = R$$

$$\text{Or, } C_P - 3R = R$$

$$\therefore C_P = 4R$$

$$\text{So the ratio of specific heats, } \gamma = \frac{C_P}{C_V} = \frac{4R}{3R} = 1.33$$

Thus the ratio of specific heat capacities of tri-atomic gas is 1.33.

But for a tri-atomic gas having 7 degrees of freedom the ratio of specific heats is 1.28.

(d) Polyatomic gas having 'f' degree of freedom:

By the law of equipartition of energy for 'f' degree of freedom we have

$$E_{Trans} = \frac{1}{2}kT + \frac{1}{2}kT + \frac{1}{2}kT + \dots \dots \dots f \text{ times} = \frac{f}{2}kT$$

Thus the energy per mole of the gas is given by, $U = N \times \frac{f}{2}kT = \frac{f}{2}RT$ [$Nk = R$]

Hence, $\frac{dU}{dT} = \frac{f}{2}R = C_V$ [Since, $C_V = \frac{dU}{dT}$]

Again we know,

$$C_p - C_V = R$$

$$\text{Or, } C_p - \frac{f}{2}R = R$$

$$\therefore C_p = \left(\frac{2+f}{2}\right)R$$

So the ratio of specific heats, $\gamma = \frac{C_p}{C_V} = \frac{\left(\frac{2+f}{2}\right)R}{\frac{f}{2}R} = \frac{2+f}{f} = 1 + \frac{2}{f}$

Thus the ratio of specific heat capacities for polyatomic gas is $\left(1 + \frac{2}{f}\right)$.

Mathematical Problems

1. Calculate [Given $N = 6.02 \times 10^{23}$ and $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$]

(a) Kinetic energy of translation of an oxygen molecule at 27 °C.

(b) The total kinetic energy of an oxygen molecule at 27 °C.

(c) The total kinetic energy of a gram molecule of oxygen at 27 °C.

- - -

Second Law of Thermodynamics

Different Statements of The Law:

Kelvin- Plank Statement

It states that practically a reservoir never gives a positive net amount of work from the heat extracted from a thermal reservoir. So, basically, you cannot have a heat engine that operates between the two temperature levels and it has no heat rejection.

It is not possible to achieve a continuous supply of work from a body by cooling it to the temperature below the coldest of its surroundings.

Clausius Statement

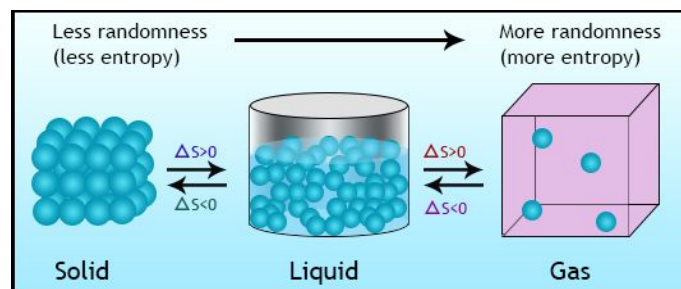
It is impossible to construct a device operating in a cycle that can transfer heat from a colder body to warmer without consuming any work. Refrigerator works on Clausius's statement. In other words, unless the compressor is driven by an external source, the refrigerator won't be able to operate.

Entropy

Entropy is a measure of the randomness of the system or it is the measure of energy or chaos within an isolated system. Entropy is a State Function because it depends only on the initial and final thermodynamic states and not on the path followed.

There are few factors that cause an increase in entropy of the closed system. Firstly, in a closed system, while the mass remains constant there is an exchange of heat with the surroundings. This change in the heat content creates a disturbance in the system thereby increasing the entropy of the system. Secondly, internal changes may occur in the movements of the molecules of the system. This leads to disturbances which further causes irreversibilities inside the system resulting in the increment of its entropy.

It is denoted by 'S'. Mathematically, $S_2 - S_1 = \int_{Q_1}^{Q_2} \frac{dQ}{T}$. Here, dQ is the energy transferred as heat, T is the temperature of the system in Kelvins. Q_1 and Q_2 stands for initial heat and final heat of the system



Entropy change of a Perfect gas

Suppose that the initial temperature, pressure and volume of m kg of a gas be T_1 , P_1 and V_1 respectively. Let this gas be heated to final temperature, pressure, volume T_2 , P_2 , V_2 respectively.

Now, according to the 1st law of thermodynamics we can write,

$$dQ = dU + dW \dots \dots \dots (i)$$

Again, we can write,

$$dW = PdV$$

$$\text{And, } dU = nC_v dT$$

Putting these two values in Eqn. (i) we get,

$$dQ = nC_v dT + PdV \dots \dots \dots (ii)$$

From ideal gas law we know,

$$PV = nRT$$

$$\text{or, } P = \frac{nRT}{V}$$

Inserting this value in Eqn. (ii) we get,

$$dQ = nC_v dT + \frac{nRT}{V} dV$$

$$\Rightarrow \frac{dQ}{T} = nC_v \frac{dT}{T} + \frac{nR}{TV} dV \quad [\text{dividing by } T]$$

Integrating,

$$\Rightarrow \int_{Q_1}^{Q_2} \frac{dQ}{T} = \int_{T_1}^{T_2} nC_v \frac{dT}{T} + \int_{V_1}^{V_2} nR \frac{dV}{V}$$

$$\Delta S = S_2 - S_1 = nC_v \ln \frac{T_2}{T_1} + nR \ln \frac{V_2}{V_1}$$

$$\Delta S = S_2 - S_1 = n \left(C_v \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1} \right)$$

This the required expression for change in entropy.

Problem: Calculate the increase in entropy of 2 gm of O₂ when its temperature is raised from 200 C to 1000 C and its volume is doubled. [C_v=25.5 Jmole⁻¹K⁻¹ and R=8.31Jmole⁻¹K⁻¹]

Difference Between Isothermal Process and Adiabatic Processes

Isothermal Process	Adiabatic Process
An isothermal process is a thermodynamic process in which there is no change in the temperature of the system. The temperature of the system remains constant throughout the thermodynamics process i.e. $\Delta T = 0$.	While an adiabatic process is the one in which there is no transfer of heat or mass in-between the system and the surrounding throughout the thermodynamic process. Hence, in an adiabatic system $\Delta Q = 0$.
From a thermal reservoir near the system, heat can be added or removed from the system, in order to keep the temperature constant.	By definition, there is no change in the heat in an adiabatic process and so there is no addition or subtraction of the heat.
Any transformation in such a process is slow.	Any transformation in such a process is fast.
Boyle's law holds good. i.e. $PV = \text{constant}$	Boyle's law doesn't hold good. i.e. $PV^\gamma = \text{constant}$

